

Vision-Language Models for Embodied Reasoning: A Comprehensive Survey

Abstract

The integration of Vision-Language Models (VLMs) into embodied agents has catalyzed a paradigm shift in robotics, transitioning from modular, task-specific pipelines to unified, generalist policies capable of complex reasoning and control. This survey provides a comprehensive analysis of the state-of-the-art in Vision-Language-Action (VLA) models and embodied reasoning systems. We synthesize findings from 300 recent academic papers to construct a detailed taxonomy of methodological families, ranging from monolithic autoregressive architectures to hierarchical dual-system frameworks. We examine critical innovations including Chain-of-Thought (CoT) reasoning adapted for physical interaction, flow-matching action generation, retrieval-augmented planning, and reinforcement learning from verifiable rewards. Furthermore, we critically evaluate the landscape of benchmarks designed to assess spatial understanding, physical commonsense, and long-horizon planning, highlighting the persistent gaps between simulation performance and real-world deployment. By analyzing limitations such as hallucination, latency, and sim-to-real transfer, we outline future research directions aimed at achieving robust, safe, and scalable embodied artificial intelligence.

1. Introduction

The quest to endow machines with the ability to perceive, reason, and act within physical environments has long been a central challenge in artificial intelligence. Historically, embodied AI systems relied on decoupled modules: perception stacks processed sensor data into symbolic representations, planners generated high-level sequences, and low-level controllers executed motor commands. While effective in constrained domains, this modular approach often suffered from error propagation and a lack of semantic grounding. The emergence of large-scale Vision-Language Models (VLMs) has fundamentally altered this landscape, offering a unified substrate for multimodal understanding that can be extended to action generation, giving rise to Vision-Language-Action (VLA) models [1], [2].

Recent advancements have demonstrated that VLMs pretrained on internet-scale data possess latent capabilities for spatial reasoning and physical commonsense that can be unlocked through specialized fine-tuning and architectural adaptations [3], [4]. Models such as PaLM-E pioneered the direct injection of continuous sensor observations into language embedding spaces, establishing the feasibility of end-to-end multimodal sentence processing [3]. Subsequent works have expanded this foundation, exploring diverse architectural strategies including mixture-of-experts (MoE) designs to disentangle understanding and control [5], [6], flow-matching decoders for precise trajectory generation [7], [8], and hierarchical frameworks that separate slow semantic reasoning from fast reactive control [9], [10].

Despite these strides, significant challenges remain. Current embodied agents often struggle with long-horizon planning, exhibiting failures in maintaining temporal consistency and causal reasoning over extended sequences [11], [12]. Furthermore, the “illusion of reasoning” persists, where models generate plausible textual rationales that are not faithfully grounded in physical constraints or visual evidence [13], [14]. The field is also grappling with the sim-to-real gap, as many high-performing models are validated exclusively in simulation environments with limited action sets or idealized physics [15], [7], [16].

This survey synthesizes the rapidly expanding literature on vision-language models for embodied

reasoning. We organize the field into coherent methodological families, analyzing the trade-offs between monolithic and hierarchical designs, the efficacy of various reasoning paradigms, and the role of reinforcement learning in aligning language priors with physical dynamics. We further provide a critical review of evaluation benchmarks, identifying key bottlenecks in assessing true embodied intelligence. By integrating insights from 300 distinct studies, this work aims to provide a definitive reference for researchers navigating the complexities of building generalist embodied agents.

2. Method

The methodological landscape of embodied reasoning with vision-language models is diverse, encompassing a spectrum of architectural choices, training paradigms, and reasoning mechanisms. We categorize these approaches into several key families: architectural foundations, reasoning paradigms, action generation mechanisms, memory and world modeling, multi-agent collaboration, and safety alignment.

2.1 Architectural Foundations of Embodied Agents

The core architecture of an embodied agent dictates how visual, linguistic, and proprioceptive information is fused to generate actions. Two primary design philosophies have emerged: monolithic end-to-end models and hierarchical dual-system architectures.

2.1.1 Monolithic End-to-End Policies Monolithic architectures integrate perception, reasoning, and action generation within a single transformer backbone, typically utilizing autoregressive decoding to predict action tokens alongside text. This approach leverages the full capacity of pretrained VLMs for unified semantic grounding.

- **Unified Tokenization and Autoregressive Decoding:** A foundational strategy involves tokenizing continuous robot actions into discrete sequences compatible with the VLM vocabulary. Methods like OpenVLA and its variants utilize byte-pair encoding (BPE) or vector quantization to map action chunks to tokens, enabling standard next-token prediction losses [1], [17]. NORA demonstrates that even smaller 3B-parameter backbones, when combined with efficient action tokenization like FAST+, can outperform larger baselines on manipulation tasks by balancing parameter efficiency with action compression [17]. Similarly, StarVLA validates the “minimal-sufficiency” hypothesis, showing that strong VLM backbones (e.g., Qwen3-VL) paired with lightweight MLP action heads can achieve state-of-the-art results without complex architectural variants [18].
- **Modality Alignment and Projection:** Effective fusion of visual and linguistic modalities is critical. PaLM-E introduced the concept of injecting continuous sensor observations directly into the LLM embedding space via learned encoders, forming multimodal sentences [3]. Recent works refine this with specialized projectors; for instance, EMAC+ utilizes a trainable linear projection layer to align frozen ViT features with LLM embeddings, facilitating bidirectional collaboration between planning and execution modules [19]. ThinkProprio advances this by discretizing continuous proprioceptive states into text tokens, allowing native integration into the VLM embedding space and outperforming learned MLP projectors [20].
- **Mixture-of-Experts (MoE) for Disentanglement:** To prevent competition between understanding and control tasks within shared parameters, several models employ MoE architectures. ChatVLA-2 integrates a dynamic MoE into the Qwen2-VL backbone, using

adaptive routing to select specific experts for multimodal understanding versus robotic control, thereby preserving open-world reasoning capabilities [5]. HY-Embodied-0.5 extends this with modality-adaptive MoE, employing non-shared QKV/FFN parameters for vision and language branches to prevent capability degradation during fine-tuning [6].

2.1.2 Hierarchical and Dual-System Architectures Hierarchical approaches decouple high-level semantic reasoning from low-level motor control, often employing a “System 2” planner and a “System 1” executor to manage latency and complexity.

- **Decoupled Planning and Execution:** The GTA-VLA model exemplifies this by decoupling slow autoregressive VLM reasoning from fast closed-loop control. It employs an asynchronous dual-system where a “Guide” phase ingests spatial priors and a “Think” phase generates structured reasoning, which then conditions a fast controller [9]. Similarly, ThinkAct employs a dual-system framework where a multimodal LLM generates embodied reasoning plans optimized via reinforcement learning, while a separate module handles low-level control [10].
- **Brain-Cerebellum Paradigms:** Inspired by biological cognition, RoboOS implements a Brain-Cerebellum hierarchy where a large MLLM (RoboBrain) handles global perception and task decomposition, while a “Cerebellum” skill library executes modular VLA tools with low latency [21]. This decoupling allows for scalable cross-embodiment collaboration via shared spatiotemporal memory.
- **Latent Bridging:** Some methods bridge the gap between high-level reasoning and low-level control using latent codes rather than explicit text. LCB (Latent Codes as Bridges) injects learnable action tokens into the LLM vocabulary to generate high-level latent goals, which are then projected to condition downstream policy networks, transmitting abstract nuances impossible to convey via natural language alone [22].

2.2 Reasoning Paradigms for Embodied Interaction

Reasoning is the cognitive engine that allows agents to decompose tasks, handle uncertainty, and adapt to dynamic environments. The adaptation of Chain-of-Thought (CoT) reasoning to the embodied domain has been a major focus.

2.2.1 Embodied Chain-of-Thought (CoT) Standard CoT reasoning involves generating intermediate textual steps before an answer. In embodied AI, this is extended to include spatial grounding, action primitives, and visual predictions.

- **Spatially Grounded CoT:** ECoT (Embodied Chain-of-Thought) augments VLA inputs with intermediate reasoning tokens that predict task rephrasing, high-level plans, and crucially, spatially grounded features like object bounding boxes and gripper positions [23]. This embedding of low-level visual grounding within the reasoning chain significantly outperforms purely textual reasoning. SpatialCoT further enforces bi-directional alignment between language-based rationales and coordinate actions, separating rationale generation from action grounding to leverage LLM strengths [24].
- **Visual CoT and Subgoal Prediction:** Moving beyond text, CoT-VLA autoregressively predicts future subgoal *images* as an intermediate reasoning step before action prediction, forcing the model to explicitly reason about desired states in pixel space [25]. HALO implements a Mixture-of-Transformers with decoupled experts for textual reasoning and visual generation, enabling simultaneous prediction of text rationales and visual subgoals [26]. VLA-Thinker

treats perception itself as a dynamically invocable reasoning action, iteratively generating textual steps and tool invocations (e.g., ZOOM-IN) before executing final actions [27].

- **Latent and Compressed Reasoning:** To mitigate the latency of autoregressive text generation, recent works explore latent reasoning. LaRA-VLA internalizes multimodal CoT into continuous latent representations, using a curriculum to transition from explicit textual CoT to compact latent states, preserving multi-step computation benefits while eliminating decoding overhead [28]. Fast-ThinkAct compresses explicit CoT into compact continuous latent vectors via preference-guided distillation, achieving significant inference speedups [29].

2.2.2 Active Perception and Uncertainty-Aware Reasoning Embodied agents must often act to gather information, a capability known as active perception.

- **Active Visual Reasoning:** PhysVLM-AVR formulates active visual reasoning as a higher-order MDP, enabling agents to identify uncertainty and predict action-conditioned observations to resolve it [30]. SceneDiver executes a coarse-to-fine focus plan, constructing scene graphs for high-level decomposition and employing agentic exploration with semantic zoom to verify details [31].
- **Uncertainty Estimation:** AdaNav integrates an Uncertainty-Adaptive Reasoning block that computes action entropy from policy outputs as an uncertainty signal, dynamically triggering deeper reasoning only when confidence is low [32]. Similarly, UPS (Uncertainty-aware Policy Steering) interleaves base policy sampling with latent world model imagination, using a VLM verifier to evaluate alignment and calibrate uncertainty via conformal prediction [33].
- **Query-Driven Perception:** RACAS reformulates detection as language-conditioned Visual Question Answering (VQA), where a controller generates task-specific visual queries and monitors return natural language descriptions, enabling zero-training adaptation to new embodiments [34].

2.2.3 Reinforcement Learning for Reasoning Alignment Supervised Fine-Tuning (SFT) alone often fails to align reasoning traces with physical outcomes. Reinforcement Learning (RL), particularly Group Relative Policy Optimization (GRPO), has become a dominant tool for optimizing reasoning policies.

- **Verifiable Rewards:** VLA-R1 and ManipLVM-R1 employ RL from Verifiable Rewards (RLVR), using metrics like Intersection-over-Union (IoU) and trajectory deviation as reward signals to bridge the gap between high-level reasoning and low-level execution [35], [36]. This approach allows models to learn physical reasoning without extensive human annotation.
- **Process Rewards:** MindPower introduces a “Mind-Reward” mechanism that aligns intermediate Belief-Desire-Intention (BDI) states with human mental states, improving action correctness in Theory-of-Mind tasks [15]. EmbodiedBrain utilizes Step-GRPO, integrating preceding planning steps as guided precursors to stabilize long-horizon policy optimization [37].
- **Self-Improvement:** EvolveNav employs a self-improving framework where formalized CoT labels are iteratively enhanced through self-reflection, reducing redundant reasoning tokens and improving inference speed [38]. SelfReVision implements a closed-loop self-critique mechanism where the model generates, revises, and verifies its own plans, enabling small VLMs to outperform much larger models [39].

2.3 Action Generation and Control Mechanisms

The translation of reasoning into physical action requires robust generation mechanisms capable of handling continuous control spaces and complex dynamics.

2.3.1 Flow Matching and Diffusion Policies Diffusion models and flow matching have emerged as superior alternatives to autoregressive token prediction for continuous action generation.

- **Flow Matching Integration:** VLASER integrates a dedicated Action Expert module utilizing flow matching for action generation, encoding robot state and noised actions as tokens that share self-attention with the language model [7]. DM0 similarly integrates a flow-matching action expert with a Qwen3 backbone, employing a hybrid gradient strategy to decouple action expert updates from VLM semantic preservation [40].
- **Discrete-to-Continuous Reconstruction:** GenieReasoner unifies reasoning and control by tokenizing continuous actions into discrete sequences via a VQ-encoder, then using a flow-matching decoder to reconstruct high-fidelity trajectories from these tokens [8]. This allows for unified autoregressive training with language tokens while recovering precise continuous control.
- **Plug-in Experts:** DexVLA integrates a billion-parameter diffusion-based action expert with a Qwen2-VL backbone, demonstrating that scaling the action expert significantly improves motor skill learning across diverse robot morphologies [41].

2.3.2 Action Tokenization and Discretization For models relying on autoregressive decoding, the method of action discretization is critical.

- **Structured Language Actions:** LAP (Language-Action Pre-training) reframes action prediction as conditional language modeling, training VLMs to auto-regressively predict structured natural language descriptions of end-effector motions, which are then parsed by a lightweight diffusion expert [42]. VLM2VLA transforms continuous robot actions into natural language descriptions to align fine-tuning data with VLM pretraining distributions, enabling parameter-efficient LoRA fine-tuning [43].
- **Spatial Action Tokens:** Embodied-R1 utilizes a unified pointing representation, generating embodiment-agnostic pointing coordinates as an intermediate step between vision-language comprehension and low-level primitives [44]. Lumo-1 employs a spatial action tokenizer that converts trajectories into discrete motion primitives via waypoint decomposition and k-means clustering [45].

2.3.3 Hierarchical and Modular Control Modular approaches allow for the integration of classical control methods with learned policies.

- **Model Predictive Control (MPC):** VLMPC integrates VLMs into an MPC loop via conditional action sampling, where the VLM generates coarse movement directions to define Gaussian sampling means for a trajectory optimizer [46]. Traj-VLMPC extends this by using VLM-conditioned Gaussian Mixture Models for trajectory-based MPC [47].
- **Code-as-Policy:** HyCodePolicy decomposes natural language into hierarchical subgoals constrained by geometric primitives, synthesizing Python programs that are executed and monitored via VLM-based visual feedback, treating code as an evolving hypothesis [48].

RoboCodeX utilizes tree-of-thought reasoning to generate executable code units with physical constraints for robotic behavior synthesis [49].

2.4 Memory, World Modeling, and Long-Horizon Planning

Handling long-horizon tasks requires agents to maintain context, model world dynamics, and plan over extended timeframes.

2.4.1 Multimodal Memory Systems Effective memory systems store and retrieve relevant spatial and semantic information.

- **Semantic-Spatial Maps:** EmbodiedLGR constructs a dual-level memory comprising a semantic graph for low-level object poses and a vector database for high-level scene captions, significantly reducing retrieval latency [50]. 3D-Mem builds compact scene memory via co-visibility clustering and frontier snapshots, enabling active exploration and lifelong learning [51].
- **Episodic and Procedural Memory:** POLAR constructs an object-centric multimodal knowledge graph linking semantic memory (user attributes) and episodic memory (trajectory summaries) for personalized navigation [52]. HELPER employs retrieval-augmented prompting with an external key-value memory of successful language-program pairs, enabling adaptation to user-specific routines [53].
- **Visual Skill Memory:** AtlasVA constructs a three-layer Visual Skill Memory hierarchy comprising spatial heatmaps, visual exemplars, and symbolic text skills, bootstrapping danger and affinity atlases directly from trajectory statistics [54].

2.4.2 World Models for Prediction and Planning World models allow agents to simulate future states and plan without interacting with the real environment.

- **Generative World Models:** WoW (World-Omniscient Worldmodel) integrates a Diffusion Transformer for video generation with a self-optimizing framework where VLM critic agents refine prompts for physical consistency [55]. COMBO factorizes joint actions for composable video diffusion modeling, accurately modeling multi-agent interactions without retraining [56].
- **Latent World Models:** GenRL connects foundation VLM embedding spaces to generative world model latent states via a noise-sampling aligner, enabling data-free policy learning by grounding language prompts into physical dynamics [57]. UNeMo integrates a Multimodal World Model based on Conditional Variational Autoencoders to predict future visual states, informing navigation decisions via a hierarchical prediction-feedback loop [58].
- **Query-Conditioned Modeling:** Physically Viable World Models argue for constructing the simplest abstraction preserving distinctions relevant to a specific intervention query, orchestrating hybrid physics engines dynamically based on the query [59].

2.4.3 Long-Horizon Planning Strategies Planning over long horizons often requires hierarchical decomposition and error recovery.

- **Hierarchical Decomposition:** HoloMind employs a two-layer planner decomposing tasks into Directed Acyclic Graphs (DAGs) for dependency management, enabling parallel execution of independent goals [60]. LaViRA decomposes navigation into three sequential stages: language action for high-level planning, vision action for grounding, and robot action for geometric control [61].

- **Error Recovery and Reflection:** PhysiAgent orchestrates a planner, monitor, and reflector, where the reflector verifies monitor outputs to generate visual constraints, creating a bidirectional feedback loop between planning and execution [62]. VLM Behavior Critic frameworks employ generator-verifier loops to identify goal-reaching yet undesirable actions, augmenting critics with external visual grounding feedback [14].

2.5 Multi-Agent Collaboration and Cross-Embodiment Learning

Scaling embodied AI often involves coordinating multiple agents or transferring skills across different robot embodiments.

2.5.1 Multi-Agent Cooperation

- **Collaborative Planning:** EMAC+ implements a bidirectional collaborative loop where an LLM generates high-level plans and a VLM executes low-level actions, allowing the planner to internalize environment-specific dynamics through retrospective feedback [19]. COMBO enables embodied multi-agent cooperation via composable world models that factorize joint actions [56].
- **Socialized Learning:** M-S²L (Multimodal Socialized Learning) combines direct RL reinforcement with multimodal observational imitation, enabling agents to learn emergent communication protocols and deictic referencing to resolve spatial ambiguities [63].
- **Centralized Coordination:** A multimodal multi-agent HRI framework employs centralized LM-based arbitration to resolve turn-taking conflicts and ensure deterministic ordering among multiple embodied agents [64].

2.5.2 Cross-Embodiment and Generalization

- **Motion Transfer:** Gemini Robotics 1.5 utilizes a Motion Transfer (MT) mechanism to align heterogeneous robot kinematics, enabling zero-shot skill transfer across distinct robot embodiments [65].
- **Robot-Agnostic Control:** RACAS encapsulates all embodiment-specific knowledge in declarative prompt configurations, enabling zero-training adaptation to new robots without modifying source code [34]. LAP represents low-level actions as natural language to foster embodiment-agnostic representations, achieving significant zero-shot transfer improvements [42].
- **Heterogeneous Training:** HEAT (Heterogeneous Embodied Agent Training) formalizes cross-morphology control as a structured POMDP, proposing Collective Adaptation as a scalable Dec-POMDP framework for distributed training [66].

2.6 Safety, Alignment, and Robustness

Ensuring the safety and robustness of embodied agents is paramount for real-world deployment.

2.6.1 Safety Constraints and Risk Mitigation

- **Contextual Safety:** CORE (Contextual Safety Reasoning and Enforcement) employs a VLM to infer context-dependent safety predicates from visual streams, mapping them to spatial safe sets via open-vocabulary segmentation and enforcing them with control barrier functions [67]. Safe-Lang integrates pre-trained VLMs to detect semantic constraints from

RGB-D images and natural language, updating failure sets dynamically for Hamilton-Jacobi reachability filtering [68].

- **Risk Identification:** HomeGuard utilizes Context-Guided Chain-of-Thought (CG-CoT) to decompose risk assessment into active perception and semantic judgment, anchoring attention to interaction targets to mitigate perceptual noise [69]. MARS (Multi-Agent Robotic System) employs a risk assessment agent that computes severity and urgency scores using hybrid risk scoring combining predefined thresholds with MLLM reasoning [70].
- **Adversarial Robustness:** ADVEDM optimizes adversarial perturbations via cross-modality embedding similarity to identify target object regions, highlighting vulnerabilities in VLM-based embodied decision-making [71]. Studies on misalignment vulnerabilities categorize four distinct interfaces (Text-Image, Text-Action, etc.) where perturbations can cause severe performance drops [72].

2.6.2 Alignment and Value Learning

- **Value Alignment:** Multimodal LLMs have been shown to act as in-context value learners, where shuffling input frames forces the model to exploit semantic grounding rather than chronological shortcuts for value estimation [73]. A value-aware vacuum cleaning framework integrates step-by-step value reasoning traces into robotic control loops to balance cleanliness and user comfort [74].
- **Preference Learning:** VLP (Vision-Language Preference Learning) defines language-conditioned preference types and constructs datasets with implicit preference orders to align policies with human preferences via offline RLHF [75]. MART (MLLM As ReTrieve) fine-tunes retrievers via preference learning on interactive feedback, prioritizing trajectory effectiveness over superficial similarity [76].

3. Challenges and Outlook

Despite significant progress, the field of vision-language models for embodied reasoning faces several critical challenges that hinder widespread deployment and true generalization.

3.1 The Illusion of Reasoning and Hallucination

A pervasive issue is the “illusion of embodied reasoning,” where models generate coherent textual rationales that are not causally linked to their actions or the physical environment. BeTTER (Benchmark for Testing True Embodied Reasoning) reveals that architectural bottlenecks, including capacity compression and myopic perceptual downsampling, degrade high-level semantic representations, leading to lexical-kinematic shortcuts rather than true understanding [13]. VLMs frequently exhibit visual grounding hallucinations, producing critiques or plans that reference non-existent objects or impossible spatial relationships [14], [77]. This disconnect is particularly evident in physical reasoning tasks, where models may possess descriptive knowledge of physics but fail to apply it procedurally during control [78], [79]. Future research must focus on developing diagnostic benchmarks and training objectives that enforce faithfulness between reasoning traces and executed actions.

3.2 Sim-to-Real Transfer and Data Scarcity

The majority of high-performing models are evaluated exclusively in simulation environments, which often lack the noise, dynamics, and complexity of the real world. Papers like MindPower, VLASER,

and ERA explicitly note the absence of real-robot experiments as a primary limitation [15], [7], [16]. The sim-to-real gap is exacerbated by the scarcity of high-quality, diverse real-world robot interaction data. While initiatives like AutoRT and InfiniteWorld aim to scale data collection through fleet orchestration and generative assets, the cost and logistical complexity of gathering real-world data remain prohibitive [80], [81]. Bridging this gap requires more robust domain randomization techniques, better physics simulators, and methods that can effectively leverage action-free internet video data for physical priors [82], [83].

3.3 Latency and Computational Efficiency

The computational demands of large VLMs pose significant challenges for real-time control. Autoregressive reasoning chains, while effective, introduce latencies ranging from seconds to tens of seconds, rendering them unsuitable for dynamic environments or high-frequency control loops [84], [85], [43]. Dual-system architectures and latent reasoning approaches offer partial solutions by decoupling slow planning from fast control or compressing reasoning into latent vectors [9], [28], [29]. However, the trade-off between reasoning depth and inference speed remains a critical design constraint. Efficient architectures, such as those utilizing intermediate layer fusion or parameter pruning, are essential for deploying generalist policies on edge devices [86], [87].

3.4 Evaluation Bottlenecks and Benchmark Limitations

Current evaluation protocols often fail to capture the full spectrum of embodied intelligence. Many benchmarks focus on short-horizon tasks or static question answering, neglecting long-horizon planning, multi-step interaction, and physical robustness [11], [88]. Metrics like success rate can be misleading if they do not account for the quality of the reasoning process or the safety of the trajectory [14], [89]. Furthermore, the lack of standardized evaluation across different embodiments and environments makes it difficult to compare methods fairly [90], [91]. There is an urgent need for comprehensive, interactive benchmarks that assess causal reasoning, physical commonsense, and adaptability in open-world settings [78], [92], [93].

3.5 Safety and Ethical Concerns

As embodied agents become more autonomous, ensuring their safety and alignment with human values becomes paramount. Current models are susceptible to adversarial attacks, misalignment vulnerabilities, and failure to recognize novel hazards [71], [72], [68]. The “black-box” nature of large models complicates the verification of safety constraints and the explanation of failure modes. Future work must prioritize the development of verifiable safety guarantees, interpretable reasoning mechanisms, and robust fallback strategies to prevent catastrophic failures in human-robot interaction scenarios [67], [70].

3.6 Future Directions

Looking ahead, several promising directions emerge. First, the integration of neuro-symbolic approaches could combine the flexibility of neural models with the rigor of symbolic reasoning, improving interpretability and robustness [94], [95]. Second, the development of “physical foundation models” that inherently understand dynamics and mechanics, rather than just semantics, is crucial for advancing physical AI [96], [97]. Third, lifelong learning and self-evolving agents that can continuously adapt to new environments and tasks without catastrophic forgetting will be essential for long-term deployment [98], [99]. Finally, fostering collaboration between the robotics and

machine learning communities to create shared infrastructure, datasets, and evaluation standards will accelerate progress toward truly generalist embodied agents.

4. Conclusion

The integration of vision-language models into embodied agents represents a transformative step towards generalist artificial intelligence capable of perceiving, reasoning, and acting in the physical world. This survey has synthesized a vast body of recent work, highlighting the diversity of architectural designs, the sophistication of emerging reasoning paradigms, and the critical importance of robust action generation and safety mechanisms. While monolithic end-to-end models offer unified semantic grounding, hierarchical and dual-system architectures provide necessary modularity and latency management. Innovations in embodied Chain-of-Thought, flow-matching control, and reinforcement learning from verifiable rewards have significantly advanced the state-of-the-art.

However, the path to robust, real-world deployment is fraught with challenges. The illusion of reasoning, the sim-to-real gap, computational latency, and safety concerns remain significant hurdles. Addressing these issues requires a concerted effort to develop more faithful reasoning mechanisms, scalable data collection strategies, efficient inference architectures, and comprehensive evaluation benchmarks. As the field matures, the convergence of large-scale foundation models with rigorous robotic control and safety principles will be key to unlocking the full potential of embodied AI. The future promises agents that not only understand the world semantically but also interact with it physically, safely, and autonomously, bridging the gap between digital intelligence and physical reality.

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